

	Assessment:	Project Milestone 1
	Subject:	Business Intelligence 381
	Total:	50

Project Milestone 1: Business and Data Understanding

Socioeconomic Development & Household Well-being in South Africa (SDHSA)

Purpose

Milestone 1 integrates the first two CRISP-DM phases. Your team must define a focused decision problem, understand the available data environment, select candidate datasets and establish evidence that the proposed project is analytically feasible.

Key principle

Do not begin by trying to use every dataset. Begin with the problem. Then investigate the data and justify which datasets are likely to help answer it.

Task 1 - Business Understanding

- Define the project background and the socioeconomic decision problem your team will investigate.
- Identify the intended stakeholder(s), their information needs and the decisions the analysis should support.
- State clear business objectives and measurable business success criteria.
- Translate the business objectives into data mining/analytics goals and technical success criteria.
- Recognize the importance of CRISP-DM as the methodology to guide the project.
- Inventory available resources, tools, skills and data assets.
- Identify risks, assumptions, constraints, dependencies and ethical concerns.
- Produce an initial project plan showing how the team will progress through the remaining milestones.

Task 2 - Data Understanding

- Inspect the full dataset portfolio before selecting subset(s) of the data for deeper investigation.
- Describe the unit of observation, dimensions, data types and important variables of candidate datasets.
- Explain the relevance of selected candidate datasets to the business problem and justify important exclusions.
- Investigate possible keys and relationships. Do not perform unexplained joins.
- Conduct and document a systematic data-quality assessment. Determine whether concerns relating to completeness, uniqueness, validity, consistency, data types,

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distributions, unusual observations, identifiers, keys and relationships are present, and support findings with appropriate R-based evidence.

- Document data quality problems
- Perform exploratory data analysis (EDA) in R using appropriate descriptive statistics and visualisations.
- Create an initial Power BI exploration showing data distributions, comparisons, correlations, initial insights and early patterns in the data. This is exploratory and is not the final dashboard.
- Formulate initial hypotheses or analytical questions that can be tested in later milestones.

Required Deliverables

- Milestone 1 Report (PDF).
- Reproducible R Markdown/Quarto notebook containing data inspection, EDA and data-quality evidence.
- Power BI file containing the initial exploratory dashboard/pages.
- Data inventory and dataset-selection matrix.
- Ethical considerations and responsible AI usage log.

Assessment Rubric

Criterion	Evidence expected	Marks
Problem background & stakeholder context	Clear, research-informed context and decision need.	5
Business objectives & success criteria	Specific objectives and measurable success criteria.	5
Data-mining goals & technical success criteria	Correct translation from business need to analytical goals.	5
Resources, risks, assumptions, constraints & ethics	Realistic project assessment including ethical risks.	5
Data inventory, selection & relevance	Evidence that multiple datasets were considered and selections justified.	7
Data description & relationship/grain analysis	Correct unit of observation, keys, relationships and limitations.	8
EDA & initial Power BI exploration	Relevant, interpretable analysis that produces initial insight.	8
Data-quality assessment	Systematic evidence of quality issues and implications.	7
TOTAL:		50

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IMPORTANT:

A claim that a dataset is “clean” or “relevant” is not sufficient. Your report must show evidence from R output, summaries, visualisations or documented inspection. You are expected to determine which quality concerns are present rather than assume that every dataset contains the same problems. Dataset relationships must be demonstrated rather than assumed.

*This project encourages originality and recognizes that there may not be a single correct answer to every aspect of the problem. Even though the emphasis is on CRISP-DM, applying the same methodology to the same business problem does not necessarily yield the same outcomes. The key is to make informed and sound decisions that align with the expectations of a data science project. The success of this project will be judged not only by the final outcome but also by the originality, rigour, logic, and thoughtfulness applied to the various stages of the project. A well-documented and well-justified project demonstrates a thorough understanding of data science principles and practices, which is a key objective of this project. **Any plagiarised work will receive a zero (0) for the project and disciplinary action will be taken.***

Additional Information

- This is a group project, but the group may not exceed three people. Groups will be allocated randomly.
- All work must be original. Copying another group’s work will not be tolerated.
- Includes names of all group members on the Cover Page.
- Submit your project milestones electronically on Moodle (BC Connect) before the due dates.
- All writing must be correctly cited and referenced.
- **Plagiarism is a serious offence.** Belgium Campus makes use of software that can scan for plagiarism and penalties apply for any plagiarised work including getting a **0** for the assignment.
- No mark will be awarded if the assignment is not uploaded via BC Connect.
- Late assignments will not be accepted; missing the deadline is an automatic 0.